

# Rakibul Islam

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raakibul.github.io

## Professional Summary

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Enthusiastic, highly motivated, who like to take initiative and seek out new challenges. My research focuses on the intersection of Machine Learning, Deep Learning, and Human-Computer Interaction (HCI). I want to design intelligent and adaptive systems that enhance how humans interact with technology by leveraging data-driven and neural network-based approaches.

**Research Interests:** Large Language Models, Machine Learning, Computer Vision, Deep Learning, Human-Computer Interaction

## Education

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*Oct 2018 – June 2023*

**University of Asia Pacific, Bangladesh**

*B.Sc. in Computer Science and Engineering*

Thesis Topic: Deep Learning-Based Background Noise Classification and Reduction for Audio Enhancement.

Relevant Coursework: Structured Programming, Data Structures and Algorithms, Object-Oriented Programming, Database, Software Engineering, Machine Learning, Artificial Intelligence, Probability & Statistics, Data Communications, Linear Algebra, Multivariable Calculus, etc.

## Thesis

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### Deep Learning-Based Background Noise Classification and Reduction for Audio Enhancement

Explanation: I worked on improving the clarity of the audio by reducing background noise. I used an LSTM-based RNN model to detect and reduce background noise in audio. This helped improve the overall clarity and quality of speech.

## Professional Experience

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*Oct 2023 – Present*

**Instructor, Caretutors Technologies Ltd.**

Delivered personalized and group-based instruction in Physics, Mathematics, and programming languages (Python, C, C++, HTML, CSS, and JavaScript for Web Development) to high school and college-level students

*Feb 2022 – May 2023*

**Industrial Trainee, Software Engineer at Brogrammers Lab**

Strengthened problem-solving skills by approaching challenges, focusing on effective solutions, and completing tasks efficiently under deadlines. Improved team collaboration and communication abilities. Developed time management skills in a professional software development environment.

*Sept 2022 – Dec 2022*

**Bdian Youth Research Trainee Program, Researcher at Youth Society for Research and Action**

Gained hands-on experience in research methodology, including proposing research topics, designing studies, and selecting appropriate methodologies for real-world applications. I also developed strong analytical and critical thinking skills through independent research, collaboration with peers, and professional presentations.

## Projects

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### Intent-Explainable Trajectory

*PyTorch, ResNet-50, LSTM, LLM (Llama-3/GPT-4)*

Designed and implemented an intent-conditioned trajectory prediction system on the PIE dataset. It generates future pedestrian trajectories conditioned on a jointly-trained crossing-intent classifier. Combining a CNN-BiLSTM encoder with an intent-conditioned LSTM decoder. Also integrated an LLM explanation layer that translates model outputs into natural-language justifications for autonomous vehicles.

### Object Detection with YOLO

*Python, PyTorch, OpenCV*

Developed as a course project. Built a real-time object detection system using the YOLO architecture to identify and classify multiple objects from images.

### Diabetes Prediction System

*Python, Pandas, NumPy, Scikit-learn, Logistic Regression*

Diabetes is a critical global health issue. A machine learningbased healthcare system that predicts the likelihood of diabetes using patient medical attributes, supporting early diagnosis and preventive care.

### Open-Source Personal Website Template for Scholars

*HTML, CSS, JavaScript, GitHub*

Designed and developed a responsive, modern, and customizable personal website template for professors to showcase research, publications, and teaching. Implemented user-friendly sections, including homepage, about, research, publications, portfolio, blog, and contact.

### Medical Insurance Cost Prediction

*Linear Regression, Polynomial Regression, Random Forest Regressor*

Developed as a course project. A predictive model that estimates medical insurance costs based on demographic and lifestyle factors such as age, BMI, smoking habits, and region.

## Technical Skills

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|-------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>ML &amp; Deep Learning</b> | Algorithms and model development                                                                                         |
| <b>Programming</b>            | Python, JavaScript, C/C++, MATLAB                                                                                        |
| <b>Packages</b>               | PyTorch, OpenCV, Pandas, Matplotlib, Keras, Scikit-learn, TensorFlow                                                     |
| <b>Web Technologies</b>       | HTML, CSS, Bootstrap, JavaScript, JSON                                                                                   |
| <b>Tools</b>                  | Microsoft Word, Excel, PowerPoint, LaTeX, Overleaf, Power BI, Git                                                        |
| <b>Project Management</b>     | Communication (Written & Verbal), Quantitative Skills, Initiative, Interpersonal Skills, Teamwork, Time Management, etc. |
| <b>Languages</b>              | English (Native, IELTS: 6.5 – L:7, W:6, R:6, S:6), Bengali (Fluent)                                                      |

## Certifications

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### 2025 – CS Fundamentals With Phitron: Spring 2025

*CS Fundamentals With Machine Learning and Deep Learning*

**2024 – Mathematics for Machine Learning and Data Science** — Luis Serrano, DeepLearning.AI  
*Linear Algebra for Machine Learning and Data Science, Calculus for Machine Learning and Data Science, Probability & Statistics for Machine Learning and Data Science* **2024 – Machine Learning Specialization** — Andrew Ng, DeepLearning.AI & Stanford University

*Supervised Machine Learning: Regression and Classification, Advanced Learning Algorithms, Unsupervised Learning, Recommenders, Reinforcement Learning*

**2024 – Learn SQL Basics for Data Science Specialization** — University of California, Davis

*SQL for Data Science by Sadie St. Lawrence*

**2024 – IBM Data Analyst Professional Certificate** — IBM

*Data Visualization with Python by Saishruthi Swaminathan and Dr. Pooja*

**2024 – Data Science Foundations Specialization — IBM**

*What is Data Science by Alex Aklson*

**2024 – Google Data Analytics Professional Certificate — Google**

*Foundations: Data, Data, Everywhere*

**2022 – Badian Research Trainee Certificate**

## References

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**Tanjina Helaly** (Advisor)

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**Dr. Alope Kumar Saha**

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**Md Shopon**

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